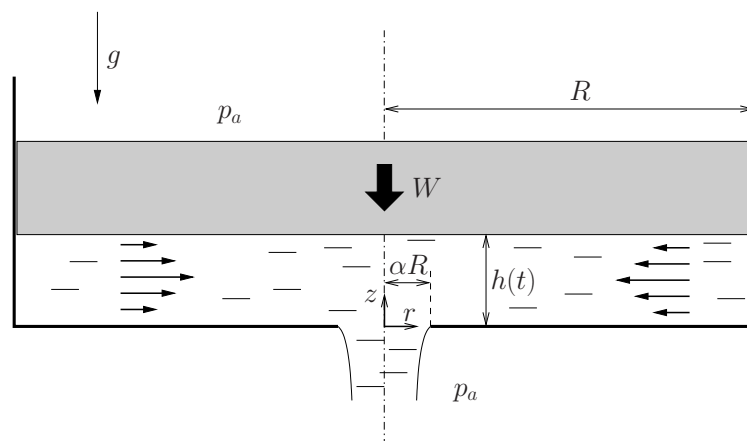


PROBLEM 4 (1 hour)

A cylindrical piston of weight W and radius R , whose upper side is exposed to the atmosphere, slides without friction inside a coaxial container of the same radius. The initial distance between the lower side of the piston and the bottom of the tank is $h_0 \ll R$. Below the piston, the container is filled with liquid of density ρ and viscosity μ . The piston falls due to its own weight, and forces the liquid to leave the tank, discharging to the atmosphere through a circular hole of radius $\alpha R \ll R$.



The known data are $(\rho, \mu, h_0, R, \alpha, W, g, p_a)$. The objective is to determine the vertical position of the piston as a function of time, $h(t)$, under the assumption of dominant viscous forces in the liquid. Gravitational forces in the liquid can be neglected ($P \approx p$). The following steps are suggested:

1. (2 points). Using the radial momentum equation, find the liquid velocity profile, $v_r(z, t)$, in the slender region $\alpha R \leq r \leq R$, $0 \leq z \leq h(t)$, as a function of $h(t)$, and of the pressure gradient $P_l = -\partial p / \partial r$.
2. (2 points). Obtain $p(r, t)$ as a function of $h(t)$, $\dot{h}(t) = dh/dt$, and known data. To that end, integrate the continuity equation across the gap (in the z direction), finding the second order differential equation satisfied by $p(r, t)$. To solve the equation, note that the pressure drop across the non-slender turning region $0 \leq r \leq \alpha R$, $0 \leq z \leq h(t)$, can be neglected, so that $p(r = \alpha R) \approx p_a$. Note also that $v_r = 0$ at the container wall $r = R$.
3. (2 points). Find the vertical force acting on the piston due to the liquid and the external atmosphere, F , as a function of $h(t)$, $\dot{h}(t)$, and known data. Note that it proves convenient to introduce a function $f(\alpha)$ to express the dependence of F on α .
4. (2 points). Using Newton's second law applied to the piston in the vertical direction, find the second-order ODE with initial conditions satisfied by $h(t)$.
5. (2 points). Obtain $h(t)$, as well as the volume flux through the hole, $Q(t)$, under the assumption of negligible inertia of the piston (acceleration of the piston much smaller than g). Note that, in this approximation, the pressure forces balance the weight of the piston, the ODE becomes of first order, and only the initial condition $h(0) = h_0$ can be satisfied.
6. (2 additional points). Provide the condition that must be accomplished for viscous forces to be dominant in the liquid, as well as the criterion of negligible inertia of the piston, in terms of the governing parameters $(\rho, \mu, h_0, R, \alpha, W, g, p_a)$.

$$\int x \ln x \, dx = \frac{x^2}{2} \left(\ln x - \frac{1}{2} \right)$$

PROBLEM 4

① DOMINANT VISCOUS FORCES : $0 = \underbrace{-\frac{\partial p}{\partial z}}_{P_L} + \mu \frac{\partial^2 v_r}{\partial z^2}$ WITH $\begin{cases} z=0: v_r=0 \\ z=h: v_r=0 \end{cases}$

GIVING POISEVILLE'S PROFILE $v_r(z, r, t) = \frac{P_L(r, t)}{2\mu} z(h(t) - z)$

② $\int_0^h \left[\frac{\partial(rv_r)}{\partial r} + \frac{\partial(rv_z)}{\partial z} \right] dz = 0 \Rightarrow \frac{\partial}{\partial r} \left[r \int_0^h v_r dz \right] + r \dot{h} = 0$

BUT $\int_0^h v_r dz = \frac{P_L(r, t) h^3}{12\mu}$ THUS $\frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) = \frac{12\mu \dot{h}}{h^3} r \Rightarrow$

$\Rightarrow r \frac{\partial p}{\partial r} = \frac{6\mu \dot{h} r^2}{h^3} + A \Rightarrow \frac{\partial p}{\partial r} = \frac{6\mu \dot{h} r}{h^3} + \frac{A}{r}$ BUT SINCE

$\left[\frac{\partial p}{\partial r} \right]_{r=R} = 0$ ($v_r(r=R)=0$) $\Rightarrow A = -\frac{6\mu \dot{h} R^2}{h^3} \Rightarrow \frac{\partial p}{\partial r} = \frac{6\mu \dot{h}}{h^3} \left(r - \frac{R^2}{r} \right)$

$\Rightarrow p = \frac{6\mu \dot{h}}{h^3} \left(\frac{r^2}{2} - R^2 \ln r \right) + B$; $[p(r=\alpha R) = p_a] \Rightarrow$

$\Rightarrow p - p_a = -\frac{6\mu \dot{h}}{h^3} \left[R^2 \ln \left(\frac{r}{\alpha R} \right) - \frac{r^2 - (\alpha R)^2}{2} \right]$

③ $F = 2\pi \int_{\alpha R}^R (p - p_a) r dr = -\frac{12\pi \mu \dot{h}}{h^3} R^4 \underbrace{\int_{\alpha}^1 \eta \left[\ln(\eta) - \ln(\alpha) - \frac{\eta^2 - \alpha^2}{2} \right] d\eta}_{f(\alpha)}$

COMPUTING THE INTEGRAL, $f(\alpha) = -\frac{1}{2} \ln(\alpha) - \frac{5}{12} + \frac{3}{4} \alpha^2 - \frac{1}{3} \alpha^3$

④ $\frac{W}{g} \ddot{h} = F - W$
 $h(0) = h_0$
 $\dot{h}(0) = 0$

⑤ IF $\ddot{h} \ll g \forall t \Rightarrow$ IN A FIRST APPROXIMATION $F = W$

WHENCE $\frac{dh}{dt} = -\frac{W}{12\pi \mu R^4 f(\alpha)} h^3$ WITH $h(0) = h_0$, WHOSE

SOLUTION IS GIVEN IN DIMENSIONLESS TERMS AS $\bar{h} = (1 + 2\tau)^{-1/2}$

WHERE WE HAVE DEFINED $\bar{h} = \frac{h}{h_0}$ AND $\tau = \frac{Wh_0^2}{12\pi \mu R^4 f(\alpha)} t$ (DIMENSIONLESS TIME)

⑥ VISCOUS FORCES DOMINANT IF $\frac{h_0 \dot{h}_0}{\mu} \ll 1$ WHERE $\dot{h}_0 \sim \frac{h_0}{t_0}$ AND $t_0 \sim \frac{\mu R^4}{Wh_0^2} \Rightarrow \frac{8Wh_0^4}{\mu^2 R^4} \ll 1$

INERTIA OF THE PISTON NEGLECTIBLE IF $\ddot{h}_0 \ll g$ BUT $\dot{h}_0 \sim \frac{h_0}{t_0} \Rightarrow \frac{W^{2/5} h_0}{\mu^2 R^8 g} \ll 1$